Detector-Free Feature Matching for Optical and SAR Images Based on a Two-Step Strategy

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***Abstract*— Optical and synthetic aperture radar (SAR) image matching presents a formidable challenge due to their pronounced geometric and radiometric distinctions arising from multimodality. The distinct imaging mechanisms of optical and SAR sensors make it challenging to identify essentially homologous points in the physical sense, raising concerns about the accuracy and repeatability of correspondences in current feature matching methods. In this study, we introduce a detector-free feature matching algorithm specifically designed to match optical and SAR images through a two-step strategy. In the initial phase, our proposed method conducts pixelwise matching (PM) using downsampled feature descriptors, eliminating the necessity to identify repeatable keypoints. To mitigate complexity, we enforce a pseudo-epipolar constraint (PEC) to reduce computational costs by constraining the search range. Subsequently, refined matching is performed on the initial correspondences to rectify inaccuracies in the PM localization of the first step. Both matching steps are implemented on a graphics processing unit (GPU) to ensure high efficiency. The proposed algorithm attains an average matching accuracy of 2.39 pixels and operates with an efficiency of 1.09 s for 1108 image pairs, underscoring its superior comprehensive performance compared to various state-of-the-art algorithms, including handcrafted methods and deep learning networks.**

***Index Terms*— Detector-free, graphics processing unit (GPU), image matching, multimodal, synthetic aperture radar (SAR).**

I. Introduction

A. Motivation

With the exponential growth in the volume of remote sensing data, addressing the processing challenges associated with big data has emerged as a pivotal concern for remote sensing processing algorithms [1]. This surge in data necessitates processing algorithms to meet stringent timeliness requirements while also catering to the demand for high real-time responsiveness in specific remote sensing applications [2]. As a foundational step for various remote sensing applications, image registration faces the emerging challenge of balancing high efficiency with sustained accuracy.

Among the various remote sensing payloads, optical and synthetic aperture radar (SAR) emerge as the most extensively utilized types. Optical sensors capture sunlight reflected from the ground, yielding images akin to the human visual system and boasting high interpretability. Conversely, SAR employs active transmission signals to generate images from radar wave signals reflected by objects. This active imaging capability enables SAR to surmount limitations encountered by optical sensors, particularly in nocturnal and cloudy conditions. Despite this advantage, SAR sensor imaging characteristics are intricate, presenting challenges for image interpretation. Acknowledging the complementary nature of these two payload types for Earth observation, the integration of optical and SAR images is common in subsequent remote sensing applications [3], [4], [5], underscoring the ongoing significance of optical and SAR image registration. However, given that optical and SAR sensors observe the Earth using different mechanisms, there is essentially no homologous point in the physical sense between optical and SAR images, leading to low keypoint repeatability even when extracted by state-of-the-art feature detectors. For instance, the multiscale block-based Shi–Tomasi (MBST) detector [6] yields a repeatability of 28.1% for a pair of optical and SAR images located at an airport, a scenario typically considered easier to match. Addressing keypoints of low repeatability may necessitate more complex algorithms and additional computational resources to ensure accurate matching, thereby incurring high computational costs and compromising matching reliability.

B. Related Work

In contrast to computer vision, remote sensing images can be roughly aligned using geographical coordinates, leading to two typical matching methods: global matching and local matching.

1) Global Matching: Global matching follows the established procedure in computer vision, involving feature detection, description, matching, and transformation estimation. These methods utilize distinct features such as keypoints, edges, and contours, carefully designed to ensure scale, rotation, and radiometric invariance, crucial for multimodal remote

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sensing images. Xiang et al. [7] proposed the optical–SAR scale-invariant feature transform (OS-SIFT), advocating for consistent gradient calculation using sensor-customized operators and the detection and matching of multiscale corners. Li et al. [8] identified keypoints based on phase congruency responses' maximum and minimum moments, constructing robust descriptors with an orientation index [radiation-variation insensitive feature transform (RIFT)]. Fan et al. [6] utilized a novel multiorientation relative total variation structural representation for feature descriptor construction, offering strong discrimination and noise robustness. Zhu et al. [10] improved keypoint repeatability using multichannel autocorrelation with log-Gabor wavelets and enhanced resistance to radiation and rotation variances by integrating the maximum index map with the DAISY structure. Fan et al. [11] focused on salient point detection in the Gaussian-scale space and developed a rotation-invariant feature descriptor via convolution with multioriented filters followed by effective data pooling operations. Xiong et al. [9] introduced a robust feature representation, adjacent self-similarity (ASS), and devised a feature detector and descriptor based on the minimum self-similarity map and the index map using ASS.

Due to their robust feature representation capabilities, deep neural networks are progressively replacing one or more modules within handcrafted global methods. For instance, Liao et al. [12] replaced handcrafted feature descriptors with a sophisticated deep convolutional neural network comprising multiple dense convolutional blocks and cross-stage partial networks. Hughes et al. [13] substituted all three modules of global matching with three deep neural networks. Xiang et al. [14] replaced feature descriptors with a pseudo-Siamese fully convolutional network and subsequently conducted matching on FAST keypoints. Quan et al. [15] introduced a deep feature network aimed at enhancing feature discrimination. Cui et al. [16] replaced feature detection and description with a convolutional network to integrate both global spatial information and local details. Xu et al. [17] proposed a local deep descriptor based on DenseNet-CSP, utilizing a Siamese network for pretraining and fine-tuning it via a pseudo-Siamese network. Hu et al. [18] proposed an improved Restormer network to generate pseudo-SAR images and adopted a point-matching-based registration network for feature matching.

2) Local Methods: Local methods in remote sensing diverge from traditional computer vision approaches by capitalizing on the geographical association between images captured by satellites. These methods rely on the precise positioning information provided by various satellites, enabling correspondence searches within predefined spatial ranges. To improve upon earlier efforts focused on similarity optimization [19], recent research has emphasized the importance of radiometric-invariant structural features, often combined with frequency-based matching techniques. Ye et al. [20] introduced a structural feature called channel features of oriented gradients (CFOGs) and conducted efficient matching in the frequency domain. Notably, deep neural networks have shown superior performance over handcrafted structural features due to their ability to learn both low-level information and high-level information. Zhang et al. [21] proposed a multilevel feature fusion method that captures both low-level fine-grained features and high-level semantic features, incorporating a feature channel excitation module for generating diverse matching features [optical and SAR image matching network (OSMNet)]. Li et al. [22] trained a neural network to predict the semantic spatial position distribution of reference and template images, eliminating the time-consuming searching process of traditional template matching. Ye et al. [23] developed an unsupervised deep neural network based on structural similarity to learn the local matching process comprehensively. In addition, Liu et al. [24] trained a feature pyramid network to extract multiscale features and transformed template matching into a regression task.

3) Global-to-Local Methods: To reconcile the limitations of both global and local methods, researchers have integrated them into a two-stage image registration framework. Ma et al. [25] initially approximated a transformation model using a convolutional neural network and then utilized more accurate features to refine the location accuracy of the initial estimation. Xiang et al. [26] proposed a two-stage algorithm for matching large-scene images, where the first stage corrects significant horizontal positioning errors, followed by a second stage refining the matching accuracy. Fan et al. [27] employed an initial estimation of an affine model using a global matching method in the coarse matching stage, followed by a template matching approach that combines log-Gabor convolution with 3-D phase correlation. Quan et al. [28] first trained a deep ordinal regression network to mitigate rotation differences and then utilized a deep descriptor learning network to address modality changes. Cao et al. [29] integrated deep feature matching based on a VGG network with dense template matching to implement a coarse-to-fine hierarchical refinement.

4) Challenges: While previous algorithms have shown promising matching results, three major challenges remain unaddressed.

1) Both global matching and local matching depend on keypoint detection. The matching accuracy of global methods is primarily contingent on the localization accuracy of repeatable keypoints, while the distribution characteristics of keypoints also have a slight impact on the accuracy of local matching. However, low keypoint repeatability between optical and SAR images obtained through different imaging mechanisms has an adverse effect on the accuracy of current methods.
2) Describing keypoints across different sensors presents a significant challenge due to adverse effects stemming from radiometric differences, geometric distortions, and severe noise. Feature descriptors of sparse keypoints require discriminative and unique characteristics, while the feature representation of local matching demands pixel-level distinguishability. This conflicting objective makes it challenging to design a feature that satisfies both global and local matching requirements.
3) To improve discrimination and distinguishability in feature representations, researchers have developed

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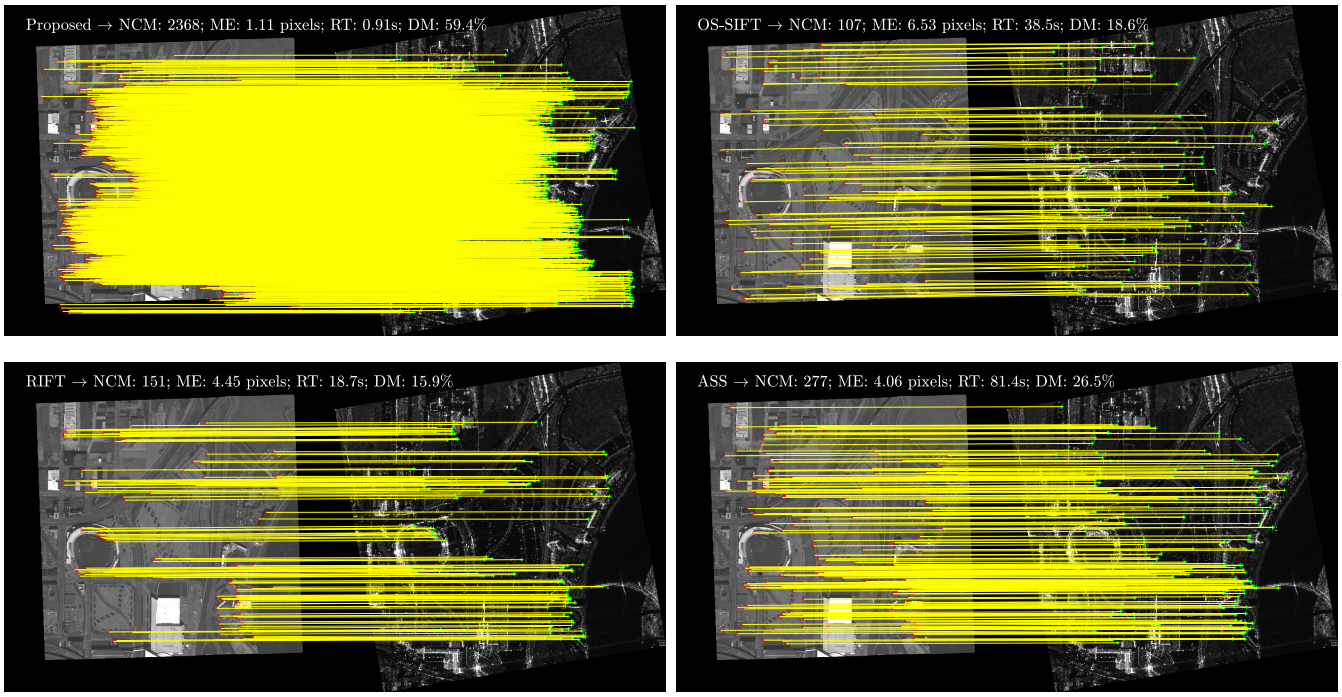


Fig. 1. Visual results of the proposed method, compared to OS-SIFT [7], RIFT [8], and ASS [9]. The proposed method excels in identifying a substantial number of high-precision correspondences with quasi-real-time efficiency (0.91 s in this instance), surpassing the performance of other methods significantly in terms of ME, RT, the NCM, and the DM.

high-dimensional feature descriptors or complex deep learning networks. However, this inevitably increases the computational burden. As illustrated in Fig. 1, OS-SIFT (136-D feature descriptor), RIFT (216-D feature descriptor), and ASS (384-D feature descriptor) exhibit considerable computational times, which are insufficient to support downstream applications with stringent processing time requirements.

C. Contribution

To address the aforementioned challenges, we introduce a detector-free feature matching algorithm aimed at achieving rapid and precise optical and SAR image registration through a two-step strategy. Initially, we perform pixelwise matching (PM) using downsampled feature descriptors, bypassing the need for detecting repeatable keypoints. Subsequently, refine matching is employed to improve the localization accuracy of these correspondences. Both matching steps are executed on a graphics processing unit (GPU) to ensure near real-time processing. The primary contributions of this study are given as follows.

1) We propose a detector-free feature matching method capable of automatically identifying sufficient and accurate correspondences in optimal areas for image matching. This approach mitigates the drawbacks associated with low repeatability and accuracy in feature detector processing of multimodal images.
2) Our methodology utilizes a two-step strategy implemented with highly parallelized GPU processing. To accelerate the PM process, we introduce a pseudo-epipolar constraint (PEC) to reduce the search range. To our knowledge, our method represents a pioneering approach in achieving high-accuracy correspondences for optical and SAR images with near real time efficiency.
3) We conduct extensive experiments on 1108 image pairs to validate the efficiency, accuracy, and reliability of the proposed method. The results demonstrate superior performance compared to state-of-the-art algorithms, including sophisticated handcrafted methods and complex deep learning networks.

D. Organization

In the following, we begin by elucidating the intricacies of the proposed method in Section II. Subsequently, Section III presents the experimental results, including comparisons with state-of-the-art algorithms. We conduct an ablation study and discuss the limitations of the proposed method in Section IV. Finally, Section V draws conclusions based on the findings of the study.

II. Methodology

In this section, we meticulously elucidate the methodology of the proposed detector-free matching method, coupled with its two-step implementation on the GPU. The comprehensive flowchart is depicted in Fig. 2.

A. Pixelwise Matching

1) Geometric Terrain Correction: Leveraging the inherent association with geographical coordinates, apparent geometric differences in remote sensing images are initially mitigated.

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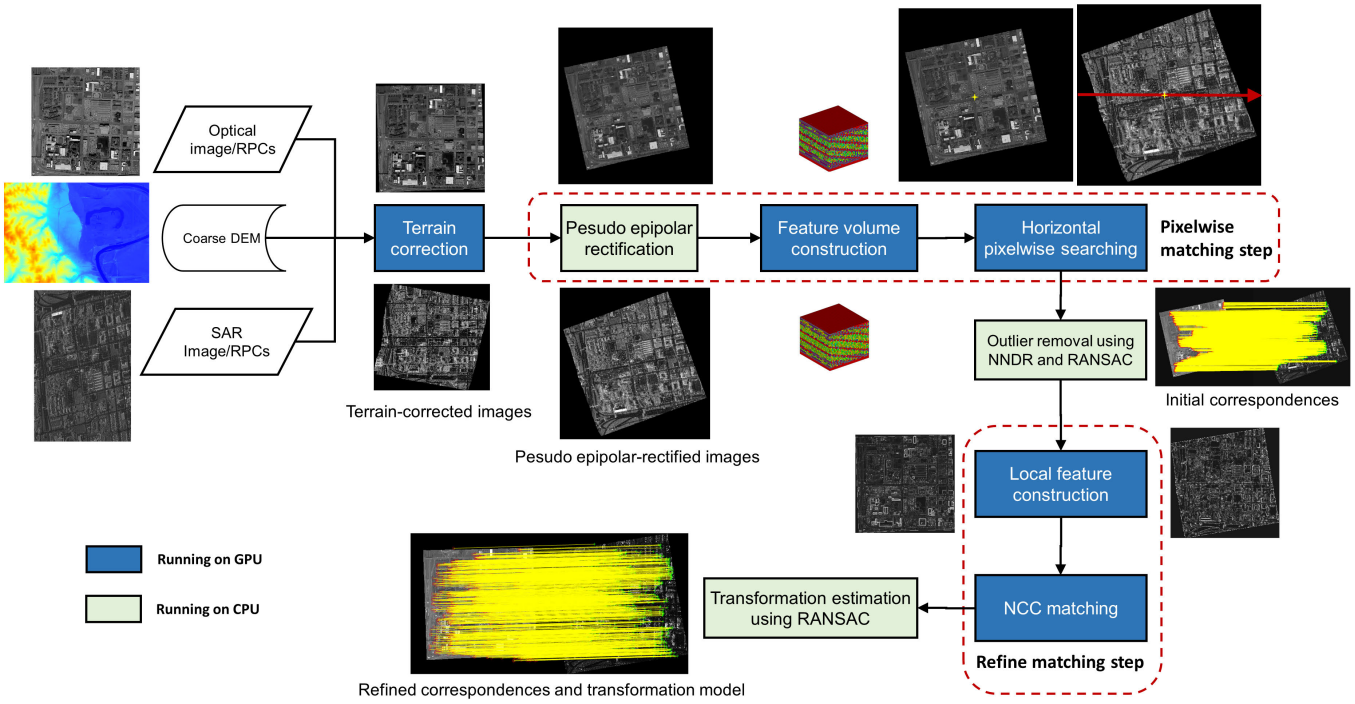


Fig. 2. Flowchart of the proposed method. We take optical and SAR images with their corresponding RPCs and a coarse DEM as inputs and then perform terrain correction to remove obvious geometric differences. PM on pseudo-epipolar-rectified images is executed to obtain numerous correspondences without feature detection, followed by the execution of refined matching to enhance the accuracy of the initial correspondences. Modules executed on the GPU are denoted in blue, while those on the CPU are denoted in green.

Given the side-looking imaging mechanism of SAR images, we initiate the process with geometric terrain correction to generate terrain-corrected images. To ensure consistency and enhance computational efficiency, we employ the versatile rational function model (RFM). This model aptly captures the intricate relationship between geographical coordinates (latitude, longitude, and altitude) and image coordinates (row and col) through two high-order polynomial equations

$$r_n = \frac{a(X_n, Y_n, Z_n)}{b(X_n, Y_n, Z_n)}, \quad c_n = \frac{e(X_n, Y_n, Z_n)}{f(X_n, Y_n, Z_n)} \tag{1}$$

where r and c represent image coordinates, and X, Y, and Z denote geographic coordinates. The variable Z is derived from a low-resolution digital elevation model (DEM). The subscript n signifies normalization, which stabilizes coordinates within the range of -1 to 1 through scale and offset factors, ensuring computational stability. Each polynomial function comprises 20 coefficients, detailed as follows:

$$\begin{aligned} p(X, Y, Z) = {} & p_0 + p_1 Z + p_2 Y + p_3 X + p_4 ZY + p_5 ZX \\ & + p_6 YX + p_7 X^2 + p_8 Y^2 + p_9 Z^2 + p_{10} ZYX \\ & + p_{11} Z^2 Y + p_{12} Z^2 X + p_{13} Y^2 Z + p_{14} Y^2 X \\ & + p_{15} ZX^2 + p_{16} YX^2 + p_{17} Z^3 + p_{18} Y^3 + p_{19} X^3 \end{aligned} \tag{2}$$

where $p \in \{a, b, e, f\}$. The geographical coordinate of each pixel in terrain-corrected images is calculated and subsequently transformed into image coordinates of slant range products. Through interpolation, these pixels are reintegrated into terrain-corrected images, a process commonly referred to as orthorectification. This rectification is particularly effective when ground control points (GCPs) and accurate digital surface models (DSMs) are employed to mitigate orbit and attitude errors. In such cases, it is imperative for object structures in orthorectified optical and SAR images to align precisely. However, the acquisition of both GCPs and DSMs is both expensive and challenging, leading to the persistence of discernible offsets in terrain-corrected images due to the inherent inaccuracies of rational polynomial coefficients (RPCs).

2) PM on Feature Volumes: To address the unresolved offsets between terrain-corrected images, global matching emerges as a preferable solution. Unlike local matching dependent on a predetermined search range, global matching explores correspondences across all feature points, offering adaptability to diverse scenarios. However, global matching confronts challenges associated with the low accuracy and repeatability of keypoints identified in optical and SAR images.

In lieu of keypoint detection and matching between two images, we advocate performing PM on downsampled feature volumes. Subsequently, correspondences, situated in the most suitable matching areas, are automatically identified. To realize this, we initially construct pixelwise feature volumes for terrain-corrected images. Given that the primary objective is to identify the most suitable matching area across the entire image, we opt for a global structural feature proposed in our previous work [30] to formulate pixelwise feature volumes, ensuring robust sensor invariance and pronounced discriminability. For terrain-corrected images, we convolve intensities with the log-Gabor filter bank to generate multiscale and

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multiorientation feature responses. Notably, the convolution with SAR images is replaced with a ratio of subwindow convolutional to overcome the interference of multiplicative speckle noise, which is expressed as follows:

$$f_{n,d}^{\text{opt}} = \sqrt{\left(I_{\text{opt}} * b_{n,d}^{e}\right)^2 + \left(I_{\text{opt}} * b_{n,d}^{o}\right)^2}$$
$$f_{n,d}^{\text{sar}} = \sqrt{\begin{array}{l}\left(\log\left(\left|I_{\text{sar}} * b_{n,d}^{e+}\right| / \left|I_{\text{sar}} * b_{n,d}^{e-}\right|\right)\right)^2 \\ +\left(\log\left(\left|I_{\text{sar}} * b_{n,d}^{o+}\right| / \left|I_{\text{sar}} * b_{n,d}^{o-}\right|\right)\right)^2\end{array}} \tag{3}$$

where d denotes the number of orientations and n signifies the number of scales. Symbol $*$ represents the convolution operation. In addition, $b_{n,d}^{e}$ and $b_{n,d}^{o}$ represent the spatial forms of the even and odd symmetric log-Gabor filter banks corresponding to the scale n and orientation d, respectively. The notation $+/-$ denotes two subconvolution windows in the spatial domain. Subsequently, we aggregate feature responses across all scales and determine the index of maximum responses over all orientations, as expressed in the following:

$$f_{\text{id}}^{\text{opt}} = \text{ind}\left(\max_{d} \sum_{n} f_{n,d}^{\text{opt}}\right), \quad f_{\text{id}}^{\text{sar}} = \text{ind}\left(\max_{d} \sum_{n} f_{n,d}^{\text{sar}}\right). \tag{4}$$

Through this operation, a consistent orientation index is calculated using filter banks with the same parameters. This ensures the maintenance of radiometric differences invariance and robustness to speckle noise, providing a solid foundation for feature description. For each pixel, we partition a 96 × 96 neighborhood centered at the pixel into 36 subpatches of 16 × 16. The histogram of the orientation index for each subpatch is concatenated to construct a 216-D descriptor, following a similar construction process as RIFT [8]. Two primary distinctions are noteworthy. First, feature responses of SAR images, calculated by ratio results, exhibit improved feature description ability and consistency with optical features. Second, we compute the orientation index of terrain-corrected images at the original resolution to preserve more details. Subsequently, pixelwise feature volumes at 1/8 of the original resolution are constructed and matched, enhancing reliability.

The subsequent process involves searching for matches in pixelwise feature volumes using the nearest neighbor distance ratio (NNDR) method employed in SIFT matching. Assuming that the sizes of terrain-corrected images are $W_1 \times H_1$ and $W_2 \times H_2$, feature volumes V_{opt} and V_{sar} are built with sizes $W_1/8 \times H_1/8$ and $W_2/8 \times H_2/8$. We search V_{sar} to find the best match for each pixel in V_{opt}. Initial matches are obtained by determining the minimum Euclidean distance of feature descriptors, followed by outlier removal.

3) Pseudo-Epipolar Constraint: While the searching process operates on downsampled feature volumes, the pixelwise calculation of Euclidean distances remains excessively time-consuming. To address this challenge, we draw inspiration from stereo matching to diminish the search range.

Serving as a pivotal step in 3-D reconstruction, stereo matching seeks pixelwise correspondences within a pair of epipolar rectified images, characterized by horizontal disparities only. Derived from the epipolar constraint of the pinhole imaging model, several researchers have proposed an approximated epipolar geometric model to adapt it to remote sensing sensors. While an exhaustive review of epipolar rectification is beyond the scope of this article, we opt to follow the method presented in [31] due to its straightforward implementation and high efficiency. This method initially orthorectifies stereo images using a coarse DSM or a constant altitude, mirroring our terrain correction process. Subsequently, the method rotates these images by an estimated angle to generate epipolar rectified images with subpixel-accuracy vertical coordinates.

The main difference lies in the fact that epipolar rectification requires bundle adjustment based on GCPs to refine the initial RPCs. In our scenario, the input RPCs of optical and SAR images still harbor noticeable errors. Despite unavoidable offsets between terrain-corrected images, a pair of optical and SAR images can be considered as a stereo observation, incorporating a PEC. Consequently, we calculate the rotation angle and transform the terrain-corrected images into pseudo-epipolar rectified images, preserving the main offset in the horizontal direction.

The outstanding issue pertains to estimating the rotation angle. Here, we propose a simple and expeditious calculation method. In theory, for two corresponding points (p_1, p_2), their geographical coordinates should be precisely aligned when GCPs and DSM are available. However, using a coarse DEM with limited accuracy in terrain correction introduces a deviation between their geographical coordinates. The direction of the deviation for all points is consistent, while the magnitude positively correlates with the DEM elevation error. Consequently, we project p_1 from the left image to p_1^t in the right image through RPCs and varying heights. Subsequently, we transform p_1^t to the terrain-corrected image coordinates

$$\begin{aligned} P_{2,j}^{t} &= G_2\left(R_2^{\text{fw}}\left(p_{1,j}^{t}, h_{\text{DEM}}\right)\right) \\ p_{1,j}^{t} &= R_2^{\text{bw}}\left(R_1^{\text{fw}}\left(p_1, h_j\right), h_j\right) \\ h_j &= \{h_{\text{DEM}} - \Delta h, \ldots, h_{\text{DEM}} + \Delta h\} \end{aligned} \tag{5}$$

where h_j denotes varying height values. R^{fw} and R^{bw} represent the forward and backward transformations based on RPCs, with the forward transformation converting image coordinates into geographical coordinates, and the backward transformation performing the reverse, respectively. G_2 signifies the transformation from geographical coordinates to terrain-corrected image coordinates. Subsequently, the terrain-corrected images undergo rotation along the line fit by the point set $P_{2,j}^{t}$, yielding pseudo-epipolar rectified images.

To enhance understanding of the proposed PEC, we provide an illustrative example highlighting its benefits. As shown in Fig. 3, a pair of terrain-corrected images is depicted, with yellow crosses representing the corresponding points and the red line indicating the search direction. Analysis of the corresponding points' coordinates reveals a reduction in vertical disparity from 101 to 9 pixels following epipolar rectification. Any remaining vertical disparity stems from RPC inaccuracies, which are negligible in our downsampled feature volumes. Unlike traditional PM, which traverses all pixels, the proposed PEC searches only along the horizontal direction, resulting in significant computational cost reduction.

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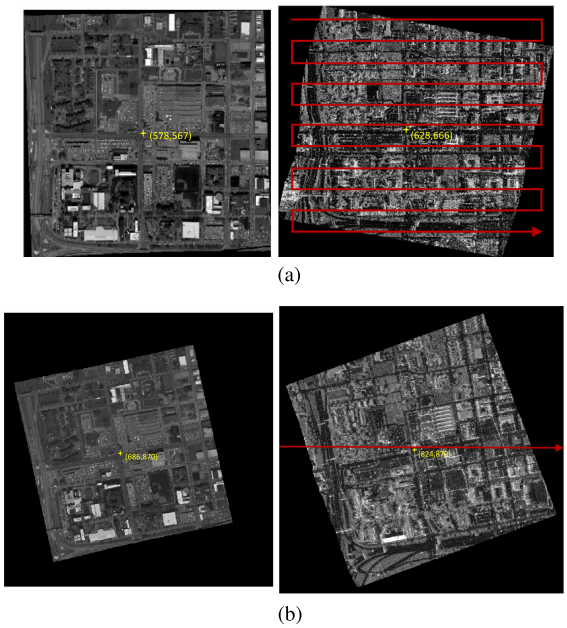


Fig. 3. Illustrative example of the PEC is presented, where yellow crosses represent corresponding points and the red line indicates the search direction. It is evident that while traditional PM requires traversal of all pixels, the proposed PEC substantially reduces computational costs. (a) Traditional pixelwise searching. (b) Proposed searching based on PEC.

The proposed search process requires only $(W_1 \times H_1 \times (W_2 \cos\varphi + H_2 \sin\varphi))/4096$ operations, with φ denoting the rotated angle. Finally, outlier removal is executed to obtain initial correspondences for the subsequent steps.

B. Refine Matching

Although the correspondences obtained in PM are situated in the most suitable regions, their localization accuracies are inherently limited due to the 1/8 scale. Consequently, we introduce refine matching to enhance accuracy further. Initially, we conduct terrain correction once more on the SAR image, refining its RPCs through a translation model estimated from initial correspondences. The remaining offset between terrain-corrected images is associated with the matching error (ME) and the downsampling rate, expressed as $8e$. To enhance fault tolerance, we set the error in PM to 2 pixels, yielding a refine matching search range of 16 pixels.

As previously mentioned, the global structural feature utilized in PM exhibits strong sensor invariance and discriminability. However, its capacity to distinguish correspondences at pixel-level accuracy is inadequate, as the global feature encodes neighborhood information to bolster robustness. Clearly, this feature is unsuitable for refine matching, where identifying the most accurate correspondence in a local range is crucial. Consequently, we employ a local structural feature and normalized cross correlation (NCC) as the matching measure. The local structural feature is an efficient adaptation of our previous work [30], relying on multidirectional anisotropic Gaussian derivatives (MAGD) to precisely and comprehensively describe intensity variation and delicate structures. We directly utilize the maximum response of MAGD among different directions as follows:

$$f_l^{\text{opt}} = \max_{\theta} \left| I_{\text{opt}} * \frac{\partial G_{\sigma,\rho,\theta}}{\partial x}_{+} - I_{\text{opt}} * \frac{\partial G_{\sigma,\rho,\theta}}{\partial x}_{-} \right|$$
$$f_l^{\text{sar}} = \max_{\theta} \left| \log\left(I_{\text{sar}} * \frac{\partial G_{\sigma,\rho,\theta}}{\partial x}_{+} \Big/ I_{\text{sar}} * \frac{\partial G_{\sigma,\rho,\theta}}{\partial x}_{-} \right) \right|. \quad (6)$$

$G_{\sigma,\rho,\theta}$ represents an anisotropic Gaussian kernel, where θ denotes the rotation angle, σ denotes the standard deviation, and ρ denotes the anisotropic factor. While f_l^{opt} and f_l^{sar} may appear similar to the previous orientation index feature in (4), the key distinction lies in the orientation index feature's accumulation of multiscale feature responses to enhance robustness. In contrast, the local structural feature relies on MAGD for a more accurate depiction of intensity variation and delicate structures. In addition, a small-scale parameter is employed to preserve pixel-level distinguishability.

Utilizing the local structural feature of a neighborhood centered at the initial correspondences as the matching objective, we employ NCC to identify the most accurate matching point within a search range of ± 16 pixels, as depicted in (7), as shown at the bottom of the next page.

(x, y) represents a pixel in the template W centered at (x_c, y_c), which denotes the image coordinates of the correspondences obtained in the PM step. The size of W is configured as $w \times w$, and $dx, dy \in [-16, +16]$. To streamline the complexity of NCC, the computation of numerators and denominators in (7) can be simplified as follows:

$$\sum_{(x,y)\in W} f_l^{\text{opt}}(x, y) f_l^{\text{sar}}(x + dx, y + dy) - w^2 \mu_{\text{opt}} \mu_{\text{sar}}^{dx,dy} \quad (8)$$

$$\sqrt{\left(\sum_{(x,y)\in W} f_l^{\text{opt}}(x, y)^2 - w^2 \left(\mu_{\text{opt}^2}\right)^2 \right) \times \left(\sum_{(x,y)\in W} f_l^{\text{sar}}(x + dx, y + dy)^2 - w^2 \left(\mu_{\text{sar}^2}^{dx,dy}\right)^2 \right)} \quad (9)$$

where μ_{opt}, μ_{opt^2}, $\mu_{\text{sar}}^{dx,dy}$, and $\mu_{\text{sar}^2}^{dx,dy}$ represent the mean feature values in the corresponding template. Notably, μ_{opt} and μ_{opt2} can be precomputed as they are independent of dx and dy. The refined matching outcomes involve identifying the highest NCC coefficient within the search range, followed by the implementation of random sample consensus (RANSAC) to achieve high-accuracy inliers and estimate a transformation model.

C. GPU Implementation

As depicted in Fig. 2, various components of the proposed method are executed on GPU. The implementation leverages the CUDA programming model, where the GPU core, streaming multiprocessors, and GPU itself are denoted as thread, block, and grid, respectively. In this model, a grid comprises blocks, and each block consists of threads.

1) Terrain Correction: Algorithm 1 outlines the pseudocode for terrain correction on GPU. The geographical extent of the terrain-corrected image is initially determined based on the forward transformation of RPCs. Subsequently, relevant

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Algorithm 1 Terrain Correction (GPU Version)

Input: Img (level-1 image product), DEM, RPCs
Output: Terrain-corrected image

Calculate the geographical transformation G
Initialize terrain-corrected image matrix Img_t
Copy Img, Img_t, G, DEM, RPCs to **GPU device**
The following part runs on GPU
Apply for 2D Block $i, j \in Img.h, Img.w$
$lon, lat = G(i, j)$, $h = DEM(lon, lat)$
$(r, c) = RPC(lon, lat, h)$, $Img_t(i, j) = Img(r, c)$
Copy Img_t to **CPU device**
End of GPU part return Img_t

data are transferred to the GPU device. Each 2-D block is responsible for processing individual pixels in the terrain-corrected image, calculating the corresponding pixel in the level-1 image, and interpolating the image intensity.

2) Pixelwise Matching: Algorithm 2 details the pseudocode for PM on GPU. Pseudo-epipolar-rectified images are first generated on the CPU by rotating terrain-corrected images to a precalculated angle. They are then convolved with filter banks on the GPU to produce multiscale and multiorientation feature responses. The orientation index feature is obtained by identifying the maximum responses of the summation on the scale. Each 2-D block on the GPU is responsible for feature calculation for each pixel. Subsequently, for each pixel on the 1/8 scale, its feature volume is constructed based on the histogram of the orientation index feature in the neighborhood. Similarly, each 2-D block is responsible for feature volume construction for each pixel. Finally, the PM similarity of two pseudo-epipolar-rectified images is calculated under supervision along the horizontal direction. This is followed by the application of NNDR and RANSAC to generate inlier correspondences.

3) Refine Matching: Algorithm 3 presents the pseudocode for refine matching on GPU. Using the initial correspondences obtained in the PM step as input, refine matching further enhances matching accuracy. A local structural feature, preserving pixel-level distinguishability, is first constructed by convolving terrain-corrected images with the first-order derivatives of multidirection anisotropic Gaussian filters. Each 2-D block on the GPU is responsible for the feature construction of each pixel in the terrain-corrected images. Subsequently, for each pair of initial correspondences, the summations of left templates are computed using the Kogge–Stone adder method to reduce the complexity of summation from N elements within $\log(N)$. Each 2-D block on the GPU is responsible for the summation of each pair of initial correspondence and a pixel in the template. Following this, the remaining calculation of NCC coefficients, requiring iteration for each offset in the search range, is performed on a 3-D block, referring to the number of initial correspondences, a pixel in the template, and an offset in the search range. The Kogge–Stone adder method is also applied in this step. Finally, the maximum NCC coefficients for each pair of initial correspondences are determined to refine the matching accuracy, followed by RANSAC to remove outliers.

Algorithm 2 PM (GPU Version)

Input: Img_e^1, Img_e^2 (pseudo epipolar-rectified images), D, N (number of orientations and scales), t_1, t_2 (thresholds of NNDR and RANSAC)
Output: Initial matches

Calculate LogGabor filter banks $b_{n,d}^e, b_{n,d}^o, n \in [1, N], d \in [1, D]$
Initialize corresponding candidates $p_1 \leftarrow (Img_e^1.w/8, Img_e^1.h/8)$, $p_2 \leftarrow (Img_e^2.w/8, Img_e^2.h/8)$
Copy Img_e^1, Img_e^2, $b_{n,d}^e$, $b_{n,d}^o$, p_1, p_2 to **GPU device**
The following part runs on GPU
Apply for 2D Block $i, j \in Img_e.h, Img_e.w$
for $d = 1$ to D **do**
  $f_d^1(i, j) \leftarrow \underset{n}{sum}(f_{d,n}^1(i, j))$
  $f_d^2(i, j) \leftarrow \underset{n}{sum}(f_{d,n}^2(i, j))$
end for
$f_{id}^1 \leftarrow ind(\max_d(f_d^1))$, $f_{id}^2 \leftarrow ind(\max_d(f_d^2))$
Apply for 2D Block $i, j \in Img_e.h, Img_e.w$
$i_s = 8i - 96$, $j_s = 8j - 96$
for $m = 0$ to 3 **do**
  for $n = 0$ to 3 **do**
    $patch \leftarrow (32m : 32m + 32, 32n : 32n + 32)$
    $seg \leftarrow (24m + 6n : 24m + 6n + 6)$
    $V_1(i_s, j_s, seg) \leftarrow histogram(f_{id}^1(patch))$
    $V_2(i_s, j_s, seg) \leftarrow histogram(f_{id}^2(patch))$
  end for
end for
Apply for 2D Block $k_1, k_2 \in (Img_e.h \times Img_e.w)/64, (Img_e.h \times Img_e.w)/64$
if $p_1(k_1).y == p_2(k_2).y$ **then**
  $dis(k_1, k_2) \leftarrow \|V_1(p_1(k_1), :) - V_2(p_2(k_2), :)\|$
end if
Sort dis along k_2, get matches and distance ratio
End of GPU part
Copy matches and distance ratio to **CPU device**
Perform NNDR and RANSAC based on t_1 and t_2 **return** coordinates of initial matches

III. Experimental Results

In this section, we initially present the experimental settings, encompassing details about the test dataset, evaluation metrics,

$$\mathrm{Ncc}(x_c, y_c, dx, dy) = \frac{\sum_{(x,y)\in W}\left(f_l^{\mathrm{opt}} - \overline{f_l^{\mathrm{opt}}}\right)\left(f_l^{\mathrm{sar}}(dx, dy) - \overline{f_l^{\mathrm{sar}}(dx, dy)}\right)}{\sqrt{\sum_{(x,y)\in W}\left(f_l^{\mathrm{opt}} - \overline{f_l^{\mathrm{opt}}}\right)^2 \sum_{(x,y)\in W}\left(f_l^{\mathrm{sar}}(dx, dy) - \overline{f_l^{\mathrm{sar}}(dx, dy)}\right)^2}} \tag{7}$$

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Algorithm 3 Refine Matching (GPU Version)

Input: Img^1, Img^2, (p_1^{im}, p_2^{im}) (initial matches), $N_{p^{im}}$ (number of the initial matches), θ_n (number of directions), w (size of template), t_3 (threshold of RANSAC)

Output: Refined matches and T (transformation model)

Calculate the first derivative of $DG_{\sigma,\rho,\theta_+}$, $DG_{\sigma,\rho,\theta_-}$
Initialize local feature f_l^1, f_l^2
Copy Img^1, Img^2, $DG_{\sigma,\rho,\theta}$, (p_1^{im}, p_2^{im}) to **GPU device**
The following part runs on GPU
Apply for 2D Block $i, j \in Img.h, Img.w$
for $d = 1$ to θ_n **do**
    $f_{l,d}^1, f_{l,d}^2 \leftarrow$ (6)
end for
$f_l^1 \leftarrow \max_d(f_{l,d}^1)$, $f_l^2 \leftarrow \max_d(f_{l,d}^2)$
Apply for 2D Block $k, t \in N_{p^{im}}, w^2$
Apply in shared memory $\mu_o[1024]$, $\mu_{o2}[1024]$, $\mu_s[1024]$, $\mu_{s2}[1024]$, $\mu_{os}[1024]$
$t_{Id} = threadId.y, b_{Id} = blockId/2$
while $t < w \times w$ **do**
    $\mu_o[t_{Id}] += f_l^1(k,t)$, $\mu_{o2}[t_{Id}] += f_l^1(k,t)^2$,
    $t += blockDim.y$
end while
while $b_{Id} != 0$ **do**
    if $t_{Id} < b_{Id}$ **then**
        $\mu_o[t_{Id}] += \mu_o[t_{Id} + b_{Id}]$,
        $\mu_{o2}[t_{Id}] += \mu_{o2}[t_{Id} + b_{Id}]$
    end if
    $b_{Id} /= 2$
end while
if $t_{Id} == 0$ **then**
    $\mu_{opt}[k] = \mu_o[0]$, $\mu_{opt2}[k] = \mu_{o2}[0]$
end if
Apply on 3D Block $k, t, q \in N_{p^{sc}}, w^2, 1089$
while $t < w \times w$ **do**
    $\mu_s[t_{Id}] += f_l^2(k,t,q)$,
    $\mu_{s2}[t_{Id}] += f_l^2(k,t,q)^2$,
    $\mu_{os}[t_{Id}] += f_l^1(k,t) f_l^2(k,t,q)$
    $t += blockDim.y$
end while
while $b_{Id} != 0$ **do**
    if $t_{Id} < b_{Id}$ **then**
        $\mu_s[t_{Id}] += \mu_s[t_{Id} + b_{Id}]$,
        $\mu_{s2}[t_{Id}] += \mu_{s2}[t_{Id} + b_{Id}]$,
        $\mu_{os}[t_{Id}] += \mu_{os}[t_{Id} + b_{Id}]$
    end if
    $b_{Id} /= 2$
end while
if $t_{Id} == 0$ **then**
    $Ncc(k,q) = \frac{(\mu_{os}[0] - \frac{\mu_{opt}[k]\mu_s[0]}{w^2})}{\sqrt{(\mu_{opt2}[k] - \frac{\mu_{opt}[k]\mu_{opt}[k]}{w^2})(\mu_{s2}[0] - \frac{\mu_s[0]\mu_s[0]}{w^2})}}$
end if
$(dx, dy)_k \leftarrow$ maximum Ncc along q
Copy (dx, dy) to **CPU device**
End of GPU part
Perform RANSAC based on t_3 **return** coordinates of refined matches and transformation model

parameter configurations, and a selection of state-of-the-art algorithms for comparative analysis. Subsequently, we provide both qualitative and quantitative results to showcase the superior matching accuracy and efficiency of our proposed algorithm.

A. Experimental Settings

1) Test Dataset and Evaluation Metrics: In our experiments, we utilize the relative orientation subset of the OS-Eval test dataset [30], encompassing 1108 pairs of optical and SAR images.[1] These images have dimensions of approximately 3500 × 3500 and 1400 × 3400 pixels. Comprehensive details of the test dataset are provided in Table I. The 1108 image pairs originate from 11 optical products and 13 SAR products situated in Omaha, USA. The optical products consist of panchromatic images from WorldView-2, WorldView-3, and SuperView-1 satellites, boasting a resolution finer than 0.5 m. On the other hand, the SAR products comprise slant range images from the GaoFen-3 (GF-3) satellite, with pixel spacings of approximately 0.56 and 0.33 m in the range and azimuth directions, respectively. Both optical and SAR products are equipped with RPCs, demonstrating an estimated accuracy better than 0.000001 pixels. The DEM with a 10-m resolution is derived from the 3D Elevation Program product available on the United States Geological Survey website. For benchmarking matching performance, ground truth points are automatically extracted using subpixel-accuracy image coordinates of metal poles in optical and SAR images, avoiding the unreliability associated with manual selection. An example of the test images is illustrated in Fig. 4. The cross structure of ground truth points is evident in SAR images, while the corresponding geometric shape and shadow are discernible in optical images, showcasing interference-free high accuracy in ground truth correspondences.

For each image pair, we estimate its transformation model T using correspondences and subsequently calculate the ME by comparing ground truth and transformed coordinates. The ME, defined in (10), is computed across all 1108 pairs

$$\mathrm{ME} = \frac{1}{N_{\mathrm{gt}}} \sum_{i=1}^{N_{\mathrm{gt}}} \left\| \left(\mathrm{gt}x_i^s, \mathrm{gt}y_i^s\right), T\left(\mathrm{gt}x_i^o, \mathrm{gt}y_i^o\right) \right\|_2 \tag{10}$$

where ME denotes the ME of an image pair and N_{gt} denotes the number of ground truths, which are described as $(\mathrm{gt}x_i^s, \mathrm{gt}y_i^s; \mathrm{gt}x_i^o, \mathrm{gt}y_i^o)$. The mean ME (MME) is then calculated as the average of MEs from all test image pairs. The second quantitative indicator is correct matching rate (CMR), which is defined as the rate of correct matching pairs under a given threshold

$$\mathrm{CMR_{th}} = 100\% \times \frac{1}{N_{\mathrm{All}}} \sum_{j=1}^{N_{\mathrm{All}}} \begin{cases} 1, & \mathrm{ME}_j < \mathrm{th} \\ 0, & \mathrm{else} \end{cases} \tag{11}$$

where N_{All} is the number of test image pairs and th denotes the threshold. We present CMR of different thresholds, including 1, 3, 5, and 10 pixels. In addition, we present the average number of correct matches (NCM), which are defined as

[1]Dataset Source: https://github.com/xym2009/OS-Eval.

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TABLE I
DETAILED DATASET DESCRIPTION

Number	Sensor	Satellite	Mode	GT	Pixel spacing	Viewing angle	Size
1108	optical	WV2-3/SV-1	PAN	28	$< 0.5m$	$2.69°\sim 23.3°$	3500×3500
1108	SAR	GF-3	Spotlight	28	$(0.56m, 0.33m)$	$18.47°\sim 37.6°$	1400×3400
1	DEM	-	-	-	10m	-	6022×2787



Fig. 4. Illustrative example of the test images, accompanied by an enlarged depiction of the ground truth points (red rectangles), is presented. As shown in yellow rectangles, the distinctive cross structure is discernible in SAR images, while the corresponding geometric shapes and shadows are evident in optical images, providing visual confirmation of the high accuracy of the ground truth data.

the inliers after outlier removal. For the sake of fairness, all the comparative algorithms adopt RANSAC of the same threshold. We also introduce the distribution of matches (DM) as a supplemental metric. Following the definition of uniform distribution outlined in [26], we divide the test image into 20×20 blocks and tally the number of blocks containing correct matches. Typically, a higher DM contributes to a more precise transformation model, consequently reducing ME. Finally, we calculated the total running time (RT) of all test images and presented the average on each test image pair. In addition to quantitative metrics, we include visual illustrations of matching point diagrams for a more comprehensive qualitative comparison.

The proposed algorithm is compared with eight state-of-the-art registration algorithms.

1) *OS-SIFT [7]:* A global matching method computes consistent gradients for optical and SAR images to construct two Harris scale spaces. Keypoints are then detected in the scale space and described using GLOH. The source code is obtained from this link.[2]
2) *RIFT [8]:* A global matching method detects corners based on the maximum and minimum moments of phase congruency responses. It builds robust descriptors using the orientation index. The source code is acquired from this link.[3]
3) *ASS [9]:* A global matching method detects feature points on the minimum self-similarity map and constructs feature descriptors based on the index map of the ASS feature representation. The encapsulated code is accessed from this link.[4]
4) *OFM [11]:* A global matching method detects feature points in the Gaussian-scale space and builds a rotation-invariant feature descriptor based on convolution with multioriented filters. The encapsulated code is obtained from this link.[5]
5) *3MRS [27]:* A global-to-local (G2L) method estimates an affine model in the first stage and then employs a template matching method to enhance matching accuracy. The source code is accessed from this link.[6]
6) *RDFM [29]:* A G2L method combines deep feature matching based on the VGG network with dense template matching, implementing a coarse-to-fine hierarchical refinement. The encapsulated code and training weights are retrieved from this link.[7]

[2]https://github.com/xym2009/OS-SIFT

[3]https://github.com/LJY-RS/RIFT-multimodal-image-matching
[4]https://gitee.com/xxin08/ASS
[5]https://github.com/Zhongli-Fan/OFM
[6]https://github.com/ZhongLi-Fan/3MRS
[7]https://github.com/Fans2017/RDFM

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TABLE II
QUANTITATIVE RESULTS OF THE COMPARATIVE ALGORITHMS ON THE TEST DATASET, COMPRISING 1108 IMAGE PAIRS. HERE, G2L SIGNIFIES GLOBAL-TO-LOCAL AND ps IS THE ABBREVIATION FOR PIXELS. HC STANDS FOR HANDCRAFTED AND DL DENOTES DEEP LEARNING. THE OPTIMAL RESULTS ARE HIGHLIGHTED IN BOLD

Category	Type	Method	MME(<10ps)	CMR_1	CMR_3	CMR_5	CMR_{10}	NCM	DM
Global	HC	OS-SIFT [7]	6.01	0.27	4.7	14.1	38.9	121	17%
Global	HC	RIFT [8]	6.34	0	0.9	12.2	48.9	162	7.4%
Global	HC	ASS [9]	4.36	1.9	29.5	59.5	92.8	235	18.9%
Global	HC	OFM [11]	4.45	1.53	26.2	56.0	88.7	88	10.2%
G2L	HC	3MRS [27]	4.57	1.89	21.8	42.6	69.6	96	9.2%
G2L	DL	RDFM [29]	5.41	0.9	10.9	22.9	51.4	482	6.3%
G2L	HC	ASS [9]+CFOG [20]	3.08	2.6	55	85.5	97.2	1215	29.2%
G2L	HC+DL	ASS [9]+OSMNet [21]	2.91	4.8	60.7	89.5	98.2	1650	41.2%
G2L	HC	Proposed	**2.39**	**5.3**	**75.1**	**96.7**	**100**	**1917**	**51.4%**

7) *ASS [9] + CFOG [20]:* A combination of ASS and CFOG, where CFOG is a local matching method estimating local offsets using a structural feature representation with fast Fourier transform (FFT) matching. The encapsulated code is obtained from this link.[8] Due to substantial positioning errors in parts of the test dataset, we combine CFOG with ASS to enhance its capability.
8) *ASS [9] + OSMNet [21]:* A combination of ASS and OSMNet, where OSMNet is a local matching method capturing both low-level information and high-level information through complex deep learning networks. The test code and training weights are obtained from this link.[9] In addition, we combine it with ASS and set its search range to 20 pixels.

We strictly follow the default configurations as implemented by the authors within their source codes. For our proposed algorithm, we have tailored the ground sampling distance during terrain correction to 1 m, considering the severe speckle noise in SAR images. In the PM step, we set the threshold of NNDR to 0.95, with the intention of retaining a broader set of potential matches while simultaneously filtering out conspicuous outliers. This is followed by the application of RANSAC, which employs a threshold of 3 pixels to validate the precision of the estimation model, thereby maximizing the retention of valid matching points, which are crucial for establishing a solid foundation for subsequent refinement processes. In the refined matching step, RANSAC is again utilized, this time with a more stringent threshold of 1 pixel, to ensure that only high-precision correspondences are maintained. The log-Gabor filter banks are configured with four scales and six orientations, aligning with the optimal settings of phase congruency. The parameters of the multidirection anisotropic Gaussian kernel are specified as $\theta_n = 9$, $\sigma = 1$, and $\rho = 2$, adhering to the widely accepted standards that facilitate the preservation of fine-grained structural features. The construction of pixelwise descriptors follows the default parameters detailed in the original RIFT paper [8], with a neighborhood size defined as 96×96. For the refinement of matches, a template size of 200×200 is employed, a parameter choice that, as discussed in numerous local matching methodologies [20], [26], offers an optimal balance between accuracy and computational efficiency.

To facilitate a fair comparison, we conduct tests on a laptop equipped with an Intel i7-11800H (16 threads@2.30 GHz) and NVIDIA GeForce RTX 3080. Under the same hardware configuration, certain comparative algorithms are executed on Windows 11, as the authors only provide encapsulated test code for this platform. In contrast, OSMNet and our proposed method are executed in Windows Subsystem for Linux2 (WSL2) with Ubuntu 18.04. In addition, we evaluate OSMNet and our proposed method, both of which can be implemented in Linux, on a server containing an Intel Xeon Silver 4216 CPU (64 threads@2.10 GHz) and NVIDIA TITAN RTX to assess their efficiency in practical applications.

B. Comparison Results

Quantitative results for the test dataset are detailed in Table II, while visual outcomes for five image pairs featuring airport, residential, suburban, and urban areas are depicted in Figs. 5 and 6. Within the comparison of algorithms, our proposed method stands out, exhibiting the smallest MME, the highest CMR, the largest NCM, and the largest DM. Specifically, the CMR achieves 96.7% under a threshold of 5 pixels, and it reaches 100% when the threshold is relaxed to 10 pixels. The MME is approximately 2.39 pixels, and our proposed method consistently maintains nearly 2000 correct matches for all test images, serving as the cornerstone for high matching accuracy. Our proposed method achieves a DM of 51.4%, indicating a notably even distribution of correspondences. Both the increased count of correct matches and the uniform distribution significantly enhance the accuracy of the transformation matrix estimated through correspondences.

[8]https://github.com/yeyuanxin110/CFOG
[9]https://github.com/zhanghan9718/OSMNet

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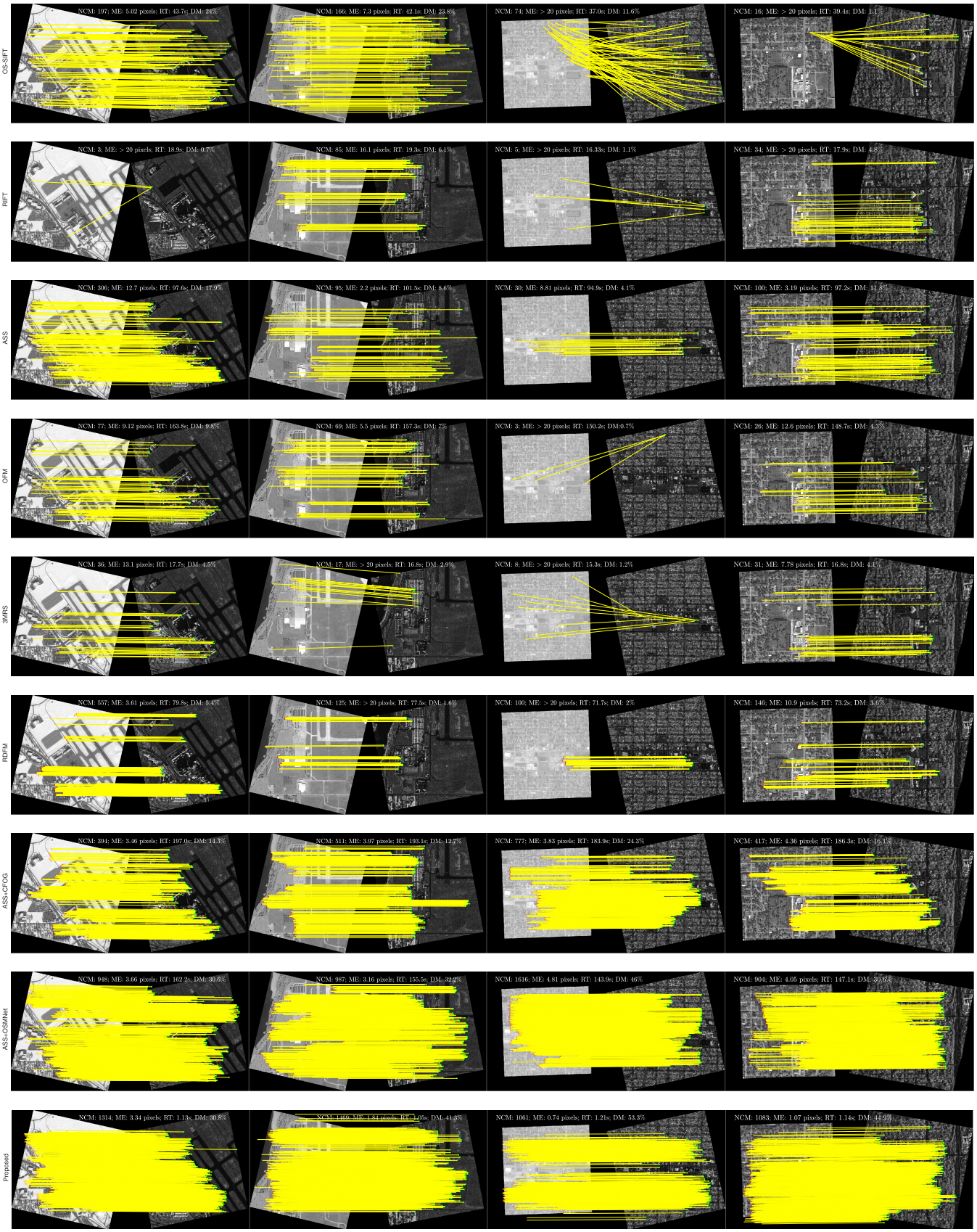


Fig. 5. Visual results of the proposed method are compared with those of eight state-of-the-art algorithms. The quantitative metrics, including NCM, ME, and RT, are also depicted to present a more detailed comparison. The proposed method demonstrates robust and efficient matching performance across various scenarios.

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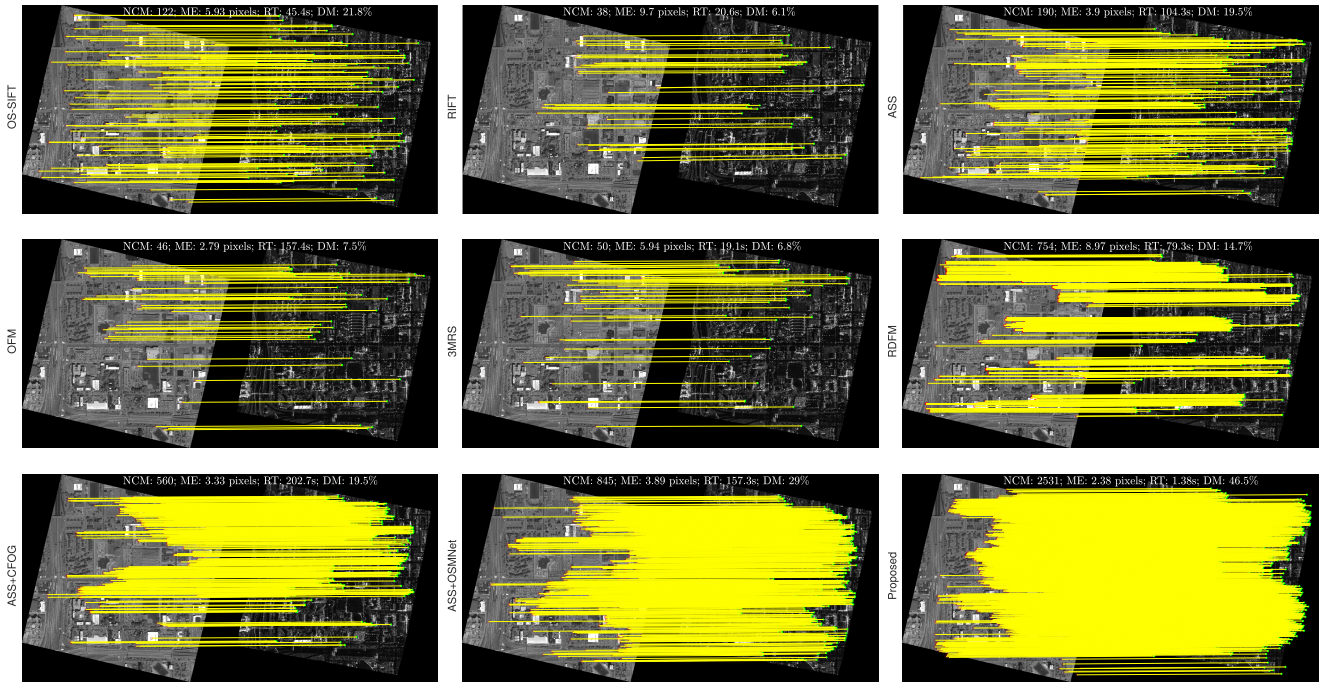


Fig. 6. Visual results of the comparative algorithms on an urban scenario. The proposed method demonstrates strong generalization ability.

In particular, the challenging scenarios illustrated in Figs. 5 and 6 are noteworthy, which are marked by substantial radiometric disparities and temporal fluctuations. In these instances, our proposed method consistently delivers superior outcomes, characterized by minimal ME and a rich set of correspondences. This performance underscores the robust multimodal matching capabilities and the method's exceptional capacity for generalization.

The advantage of searching correspondences from all candidate keypoints is evident in the high reliability demonstrated by global matching methods, particularly on image pairs with substantial positioning errors. Among these methods, ASS stands out, achieving an MME of 4.36 pixels, a CMR_{10} of 92.8%, an NCM of 235, and a DM of 18.9%, indicating high reliability and acceptable accuracy. However, the low localization accuracy and keypoint repeatability straightforwardly weaken the matching performance of global methods, resulting in a relatively low NCM and introducing ambiguity to the estimation of the transformation model. OS-SIFT, focusing on consistent gradient features, produces acceptable results in areas rich in structural information, such as airports, but struggles with matching images in residential areas. RIFT's correspondences concentrate on large building structures, hindering the accurate estimation of a transformation model due to the uneven distribution of matching points. OFM, the second-best global method, achieves an MME of 4.45 pixels and a CMR_{10} of 88.7%. However, it falters in the residential case shown in Fig. 5 due to the low repeatability of its feature detector.

G2L algorithms yield superior results compared to global methods by incorporating local matching to enhance matching accuracy. Typically, these algorithms commence by estimating a transformation model between reference and sensed images using global correspondences, followed by warping the sensed image based on the obtained model. Subsequently, local correspondences are determined from the reference and warped images, employing uniformly distributed keypoints and local matching techniques.

However, existing G2L algorithms confront two significant challenges. First, the reliability of global matching is crucial to ensuring overall matching accuracy. Failure in establishing a dependable transformation model during the global matching step diminishes the likelihood of correct matching points during subsequent local matching. Among G2L algorithms, ASS [9] serves as the global matching method for ASS + CFOG and ASS + OSMNet. Its high matching reliability, CMR_{10} of 92.8% and CMR_{20} of 98.3%, provides a solid foundation. Consequently, both local matching methods, CFOG and OSMNet, exhibit high accuracy based on this foundation. However, for two other G2L algorithms, 3MRS uses a simplified version of RIFT as the global matching method, suffering from low keypoint repeatability. RDFM utilizes dense matching of deep features extracted by the pretrained VGG16 network, struggling to handle significant differences between high-resolution optical and SAR images of complex areas. Conversely, while RDFM yields a high NCM, its correspondences tend to cluster in localized regions, leading to inaccuracies in the transformation models. This observation is corroborated in Figs. 5 and 6. Second, the local matching step in existing G2L algorithms typically extracts uniformly distributed keypoints in the optical image and then searches for correspondences in a small range of the SAR image. A high uniform distribution of final correspondences enhances the accuracy of the estimated transformation model. The block-Harris method [20], which divides the optical image into several blocks and detects the most significant corners in each block, is the most common approach. However, optical corners may not have accurate corresponding points in SAR

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TABLE III
AVERAGE RTS OF VARIOUS ALGORITHMS ARE COMPARED ON A LAPTOP. IN ADDITION, ASS + OSMNET AND THE PROPOSED METHOD UNDERGO FURTHER TESTING ON A LINUX SERVER TO EVALUATE THEIR PERFORMANCE IN PRACTICAL APPLICATIONS. THE ASTERISK (*) INDICATES THAT THE ALGORITHM IS EXCLUSIVELY TESTED IN A WINDOWS ENVIRONMENT. HC STANDS FOR HANDCRAFTED AND DL DENOTES DEEP LEARNING

Category	Type	Method	RT-laptop(s)	RT-server(s)	Implementation	With GPU
Global	HC	OS-SIFT	40.6	*	Matlab	no
Global	HC	RIFT	18.9	*	Matlab	no
Global	HC	ASS	100.5	*	Matlab-package	no
Global	HC	OFM	160.5	*	Matlab-package	no
G2L	HC	3MRS	18.5	*	Matlab	no
G2L	DL	RDFM	76.6	*	Matlab-package	yes
G2L	HC	ASS+CFOG	194.4	*	Matlab&C++mex	no
G2L	HC+DL	ASS+OSMNet	155.1	125.4	Matlab&Pytorch	yes
G2L	HC	Proposed	**1.93**	**1.09**	C++	yes

images due to imaging mechanisms, making it challenging to obtain uniformly distributed correspondences through local matching on optical corners.

In addition to evaluating matching accuracy, we present the matching efficiency of each algorithm on the test dataset. Table III details the average RT for 1108 image pairs. Under the same hardware configuration on a laptop, the proposed method requires only 1.93 s per image pair, demonstrating an acceleration of approximately tenfold compared to 3MRS, the second-fastest algorithm, and roughly 80-fold compared to ASS + OSMNet, the second most accurate algorithm. Existing algorithms, primarily focused on designing sophisticated features to address significant differences between optical and SAR images, tend to overlook computational efficiency in the challenging task of optical and SAR image registration. In contrast, we proposed a highly parallelized method, which leverages GPU implementation and a PEC to limit the search range, resulting in a significant enhancement in matching efficiency. Subsequent results obtained on the server reveal that the proposed method takes only 1.09 s on the Linux server. The RT of the proposed method encompasses the entire process from loading the input images to saving the coordinates of matching points, providing an accurate and efficient foundation for downstream applications relying on optical and SAR image registration.

IV. DISCUSSION

In this section, we initiate a comprehensive experiment to investigate the modules within our two-step strategy that have significantly enhanced both matching efficiency and accuracy. Subsequently, we discuss the limitations of the proposed method, providing insights to guide our future research endeavors.

A. Ablation Study

The contributions of the proposed method are delineated across four dimensions. First, in the PM step, correspondences are directly sought from the nearest neighbors of all candidates, eschewing the traditional feature detection and matching approach. This mitigates the challenge of low keypoint repeatability. Second, a PEC is introduced to confine the search range in PM, thereby expediting the search process. Third, the refine matching step enhances the localization accuracy of initial correspondences acquired through PM, bypassing the need for block-Harris corner detection. This refinement results in the acquisition of numerous inliers, facilitating the estimation of an accurate transformation model. Finally, both pixelwise and refine matching steps are optimized for GPU acceleration. Experiments are conducted, isolating the contributions to assess their individual impacts on the overall improvement of the proposed method.

The results of ablation experiments are detailed in Table IV. The proposed method w/o PM, wherein PM is substituted with our previously introduced global method [30], manifests the highest degree of accuracy loss. Traditional global matching methods, grappling with pronounced distinctions and distinct imaging mechanisms between optical and SAR images, falter in extracting adequate repeatable keypoints for estimating a precise transformation model. In contrast, our proposed PM method explores global correspondences across all pixels, leveraging features with high-level information and a broader receptive field, thereby enhancing adaptability in challenging scenarios. The proposed method w/o PEC, which discards PEC to delimit the search range, yields the second-worst results. PM, searching correspondences across all pixels, may encounter ambiguous matches with similar similarity, leading to local optima that fail the NNDR test, resulting in a decline in inliers. PEC not only aids in focusing the search process on relevant candidates but also reduces calculations from $(W_1 \times H_1 \times W_2 \times H_2)/4096$ to $(W_1 \times H_1 \times (W_2 \cos\varphi + H_2 \sin\varphi))/4096$. The proposed method w/o IC, which employs the block-Harris method for corner extraction rather than

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TABLE IV
ABLATION EXPERIMENTS TO INVESTIGATE THE PERFORMANCE IMPROVEMENT OF EACH CONTRIBUTION. W/O SIGNIFIES WITHOUT, PM DENOTES PIXELWISE MATCHING, PEC STANDS FOR PSEUDO-EPIPOLAR CONSTRAINT, AND IC REFERS TO INITIAL CORRESPONDENCES

Method	MME	CMR_1	CMR_3	CMR_5	CMR_{10}	NCM	DM	RT-server(s)
Proposed	**2.39**	**5.3**	**75.1**	**96.7**	**100**	**1917**	**51.4%**	**1.09**
Proposed w/o PM	2.79	2.17	59.2	90.2	99.5	1142	26.6%	4.85
Proposed w/o PEC	2.65	3.43	62.5	93.4	99.6	873	23.5%	1.72
Proposed w/o IC	2.47	4.87	72.8	94.9	98.8	1498	40.7%	1.37
Proposed w/o GPU	2.42	5.1	75	96.1	99.9	1330	45.8%	51.6

refining initial correspondences from PM, exhibits a modest decline in both matching accuracy and efficiency. Although uniformly distributed corners in the optical image can be detected using block-Harris, there is no assurance that inliers derived from corners still possess a sufficient number and good distribution characteristics for the subsequent transformation estimation. The proposed method w/o GPU, which executes the method on CPU, demonstrates the lengthiest RT. Our GPU implementation accelerates the process by 47 times compared to the CPU implementation. While the matching accuracy of the proposed method without GPU remains comparable to the original version, slight disparities arise from distinct NCC implementations, with the GPU version directly following (7) and the CPU version utilizing FFT to expedite calculations. In summary, all the aforementioned contributions yield a substantial enhancement in matching accuracy and efficiency.

B. Limitation and Future Work

In this article, we present a detector-free feature matching method tailored for high-resolution optical and SAR images. While the method demonstrates commendable performance in both matching accuracy and efficiency, it exhibits several limitations.

1) *Limitation of GPU Graphics Memory:* Our detector-free method involves traversing all pixels during the search process, theoretically consuming time. Although we mitigate this by introducing a PEC to reduce the search range and leveraging GPU acceleration for calculations, the test images must be directly transferred to the graphics memory for pixelwise feature construction and matching on the GPU. Given the size of the test image pairs (approximately 3500 × 3500 and 1400 × 3400 pixels), nearly 6 GB of graphics memory is required for the PM step. Our future work will focus on addressing this limitation by exploring two approaches: breaking down extensive remote sensing images into smaller patches and striking an optimal balance between memory constraints and parallel efficiency.
2) *Gap With Real-Time Applications:* Despite achieving efficient and effective results compared to other algorithms, the proposed method still requires nearly 1 s to process a pair of test images of size 3500 × 3500 and 1400 × 3400 pixels, posing a significant obstacle for real-time applications. Certain segments of the method are still executed on the CPU, necessitating data transfers between host and device memories. In future work, we plan to investigate a fully GPU-centric implementation to reduce computation times. In addition, precomputing and reusing specific calculations for neighboring computations can minimize data transfers between the global memory of the GPU, thereby enhancing processing speed.
3) *Singularity of Testing Data Sources:* Currently, the proposed method primarily focuses on panchromatic images for optical sensors and C-band images for SAR sensors. Since our method relies on sensor-invariant structural features and a two-step matching framework, it demonstrates robustness against differences in optical and SAR data across various bands. Future experiments will explore the generalization of our method by testing on multispectral bands of optical sensors and X-/Ku-/Ka-/L-bands of SAR sensors.

V. CONCLUSION

This article introduces an efficient and accurate matching method tailored for high-resolution optical and SAR images. The method adopts a two-step strategy. In the initial step, PM is performed to acquire a substantial number of initial correspondences situated in the most suitable matching areas. The subsequent step refines the accuracy of these initial correspondences to deduce a precise transformation model. Comparative evaluations against eight state-of-the-art algorithms demonstrate the superior matching performance of the proposed method in terms of both accuracy and efficiency. The primary advantages of the proposed method are summarized as follows.

1) The proposed method diverges from the conventional feature detection and matching process by automatically identifying initial correspondences in the most similar areas using high-level features, followed by a refinement of localization accuracy.
2) Leveraging a PEC to limit the search range, the proposed method employs a two-step matching approach on the GPU, achieving quasi-real-time computational efficiency.

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